

- d) It is not safe to hike on the trail, but grizzly bears have not been seen in the area and the berries along the trail are ripe — $\neg (79 \wedge (7P \wedge 7Q))$
- e) For hiking on the trail to be safe, it is necessary but not sufficient that berries not be ripe along the trail and for grizzly bears not to have been seen in the area — $\neg (7P \wedge 7Q) \rightarrow 79$ $\wedge$ $Q \rightarrow 7(7P \wedge 7Q)$
- f) Hiking is not safe on the trail whenever grizzly bears have been seen in the area and berries are ripe along the trail — $(P \wedge Q) \rightarrow 79$

24. Write each of these statements in the form "if p, then q" in English

- a) It is necessary to wash the boss's car to get promoted.
 Ans: If somebody wants to get promoted, then that person must wash the boss's car.
- b) Winds from the south imply a spring thaw.
 Ans: If winds are coming from the south, then a spring thaw will occur.
- c) A sufficient condition for the warranty to be good is that you bought the computer less than a year ago.
 Ans: If you bought the computer less than a year ago, then the warranty must be good.
- d) Willy gets caught whenever he cheats.
 Ans: If Willy cheats, then he gets caught.
- e) You can access the website only if you pay a subscription fee.
 Ans: If you can access the website, then you must have paid a subscription fee.
- f) Getting elected follows from knowing the right people.
 Ans: If someone knows the right people, then that person can get elected.
- g) Carol gets seasick whenever she is on a boat.
 Ans: If Carol is on a boat, then she gets seasick.